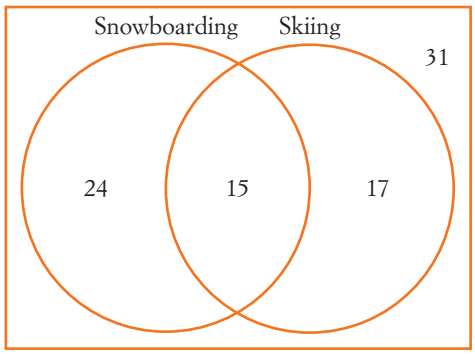
Name: _____

WORKSHEET

Venn diagrams

1 A group of students were asked if they had ever been snowboarding or skiing. The results are shown in the Venn diagram below.



a How many students have been snowboarding?

b How many students have been skiing or snowboarding?

c How many students have been snowboarding and skiing?

d How many students have been skiing but not snowboarding?

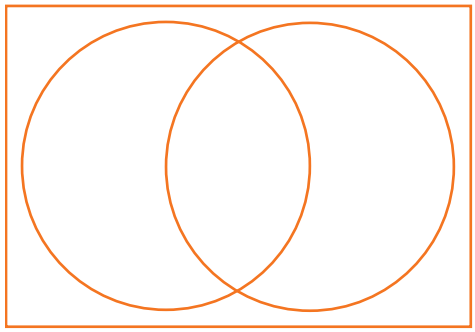
e How many students have never been skiing or snowboarding?

f How many students have been skiing or snowboarding, but not both?

g How many students have not been skiing and snowboarding?

h What is the total number of students represented in the Venn diagram?

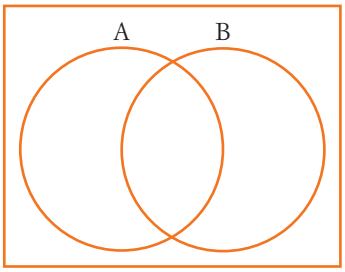
2 A group of 110 Year 11 students were asked about the computers they had at home. There were 58 students who had laptops at home, whereas 45 had a desktop and no laptop. There were 18 students who had both a laptop and a desktop at home. Display this information on the Venn diagram below.



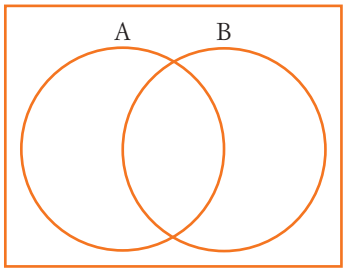
© Cengage Learning Australia Pty Ltd 2013

3 Shade in the following regions.

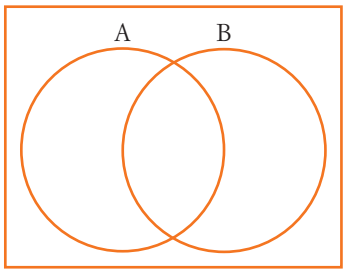
a $A \cup B$



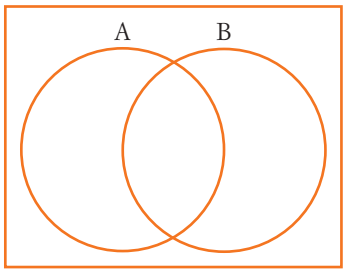
b $A \cap B$



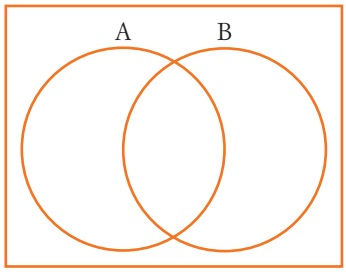
c $A \cap \overline{B}$



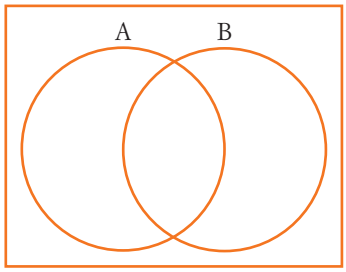
d $\overline{A \cup B}$



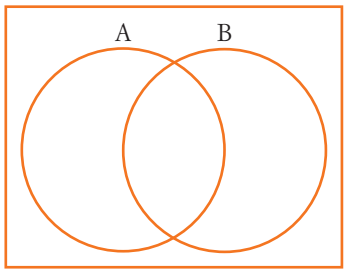
e The universal set.



f $\overline{A}$

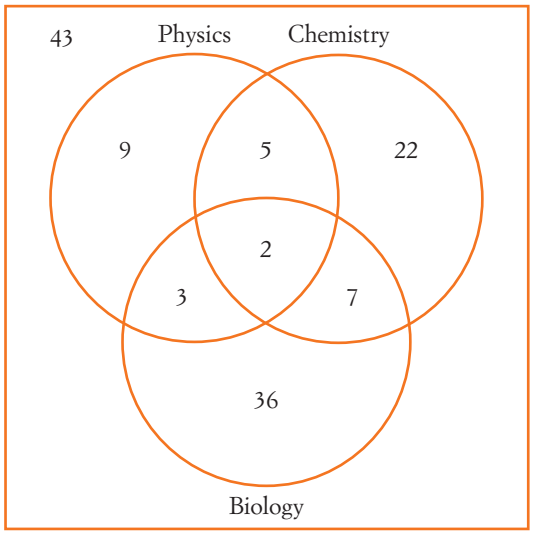


g $\overline{A \cap \overline{B}}$



© Cengage Learning Australia Pty Ltd 2013

4 Students were asked about the subjects they were studying at school. The information was collected and displayed in the Venn diagram below.



Given that Physics $= P$, Chemistry $= C$, and Biology $= B$, how many students belong to:

a $P \cup C$

b $P \cap C$

c $(P \cap C) \cup B$

d $(P \cap C) \cup \overline{B}$

e $P \cap C \cap B$

f $P \cap \overline{C} \cap B$

g $\overline{P} \cap \overline{B \cap C}$

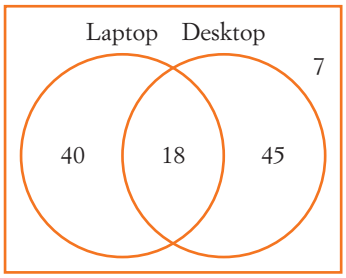
h $P \cap \overline{C} \cap \overline{B}$

© Cengage Learning Australia Pty Ltd 2013

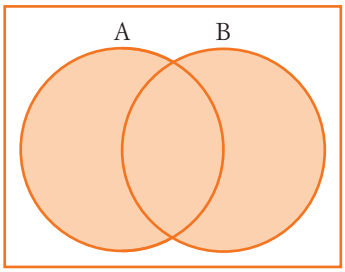
Answers

1 **a** 39
b 56
c 15
d 17
e 31
f 41
g 72
h 87

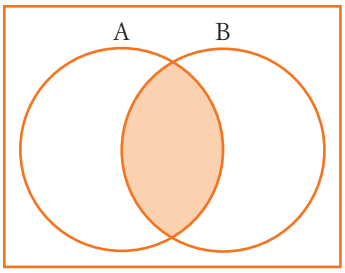
2



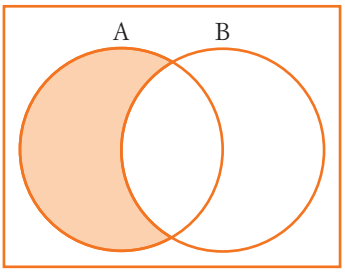
3 **a**



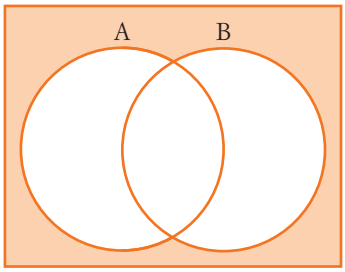
b



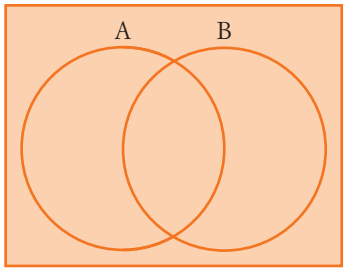
c



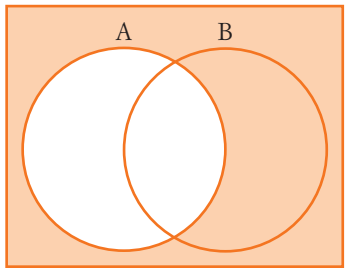
d



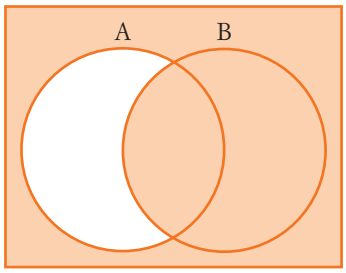
e



f



g



4 **a** 48
b 7
c 53
d 81
e 2
f 3
g 101
h 9

© Cengage Learning Australia Pty Ltd 2013